



Science — Cbse Annual Exam — 80 Marks

ANSWER KEY — FOR GRADING

Subject: Science | Class: Class 8

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	How do particles behave in materials that are gases compared to solids? (A) They cannot move much and are fixed in position (B) They are bonded together to form pure substances (C) They combine physically to form mixtures only (D) They can move around freely Answer: Correct Option: (D) ► Why? D. They can move around freely because the text states that in gases particles can move around freely, while in solids they cannot move much. [Ch 1: Exploring the Investigative World of Science]	1
2	What does the image of a root on the left-hand pages of the textbook symbolize? (A) The need to fly high in scientific exploration (B) The starting point of all experiments (C) The deep, solid foundation of knowledge connected to environment, traditions, and heritage (D) The importance of plants in science Answer: Correct Option: (C) ► Why? C. The deep, solid foundation of knowledge connected to environment, traditions, and heritage. The text states that the root symbolizes the deep, solid foundation of knowledge that keeps us connected to our environment, traditions, and our cultural and natural heritage. [Ch 1: Exploring the Investigative World of Science]	1



Q.NO.	QUESTIONS	MARKS
3	Assertion (A): Human-made ecosystems like agricultural fields must be made sustainable. Reason (R): Agricultural fields are necessary ecosystems. (A) A is true but R is false. (B) Both A and R are true, and R is the correct explanation of A. (C) Both A and R are true, but R is NOT the correct explanation of A. (D) A is false but R is true. Answer: Correct Option: (C) ► Why? C. Both statements are true according to the source which states 'Human-made ecosystems like agricultural fields are necessary, but they must be made sustainable.' The reason states that agricultural fields are necessary but does not explain why they must be sustainable—these are two separate points from the original statement. [Ch 12: How Nature Works in Harmony]	1
4	Which of the following forms a non-uniform mixture when mixed with water? (A) Chalk powder (B) Sugar (C) Salt (D) Oxygen gas Answer: Correct Option: (A) ► Why? A. Chalk powder — When chalk powder is mixed with water, the components are not evenly distributed, forming a non-uniform mixture. Salt, sugar, and oxygen dissolve evenly in water to form uniform mixtures. [Ch 9: The Amazing World of Solutes, Solvents, and Solutions]	1
5	When a bicycle stops pedalling on a flat road, the bicycle slows down and comes to a stop. What causes this change in speed? (A) A force is acting on the bicycle to bring it to rest (B) No force acts on the bicycle (C) The gravitational force pulls it backward (D) The muscular force of the rider continues to act Answer: Correct Option: (A) ► Why? A. A force is acting on the bicycle to bring it to rest because a force is essential to change the speed of an object, and the bicycle slows down due to a force acting on it. [Ch 5: Exploring Forces]	1



Q.NO.	QUESTIONS	MARKS
6	When the ends of the cylindrical coil are connected to the terminals of an electric cell, what happens to the magnetic compass needles placed near the ends of the coil? (A) They remain completely stationary (B) They point vertically upward (C) They show deflection from their original positions (D) They lose their magnetic properties Answer: Correct Option: (C) ► Why? C. They show deflection from their original positions because connecting the coil to the electric cell produces a magnetic effect that causes the compass needles to deflect. <i>[Ch 4: Electricity: Magnetic and Heating Effects]</i>	1
7	What does the syringe experiment demonstrate about gas particles when the plunger is pushed inward? (A) Gas particles have space between them that can be reduced by applying external pressure (B) Gas particles disappear when compressed (C) Gas particles change into liquid particles when compressed (D) Gas particles have no space between them Answer: Correct Option: (A) ► Why? A. The experiment shows that gas particles have a lot of space between them in their natural state, and this space can be reduced by applying external pressure. <i>[Ch 7: Particulate Nature of Matter]</i>	1
8	Dr. Ananda Mohan Chakrabarty developed a bacterium in 1971 that could: (A) Decompose plastic waste (B) Break down oil spills (C) Ferment flour for cooking (D) Produce biogas from household waste Answer: Correct Option: (B) ► Why? B. Break down oil spills. Dr. Chakrabarty developed a special bacterium in 1971 that could break down oil spills, helping to clean the environment. <i>[Ch 2: The Invisible Living World: Beyond Our Naked Eye]</i>	1
9	Why are overhead water tanks placed at a height in buildings? (A) To make them easier to fill with water (B) To increase pressure in the taps, resulting in a good stream of water (C) To protect them from damage at ground level (D) To reduce the cost of construction (E) To reduce evaporation of stored water	1



Q.NO.	QUESTIONS	MARKS
	Answer: Correct Option: (B) ► Why? B. Overhead tanks are placed at a height so that the pressure in the taps is increased, resulting in a good stream of water from the taps. [Ch 6: Pressure, Winds, Storms, and Cyclones]	
10	By how much does Makar Sankranti move ahead in the tropical calendar? (A) One day every 50 years (B) One day every 71 years (C) One day every year (D) One day every 100 years Answer: Correct Option: (B) ► Why? B. One day every 71 years. Due to the slow wobble of Earth's axis, Makar Sankranti moves ahead by one day every 71 years. [Ch 11: Keeping Time with the Skies]	1
11	Assertion (A): The range of distances from a star where water remains liquid is called the habitable zone. Reason (R): The habitable zone is also known as the Goldilocks zone. (A) A is false but R is true. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) Both A and R are true, and R is the correct explanation of A. Answer: Correct Option: (B) ► Why? B. Both statements are true as the habitable zone is defined as the range where water remains liquid, and it is also called the Goldilocks zone. However, the reason only provides an alternate name and does not explain the definition given in the assertion. [Ch 13: Our Home: Earth, a Unique Life Sustaining Planet]	1
12	The human eye contains which type of lens inside it? (A) Concave lens (B) Plane mirror (C) Convex lens (D) Flat glass plate Answer: Correct Option: (C) ► Why? C. Convex lens because even our eye has a convex lens inside it that can change its shape to allow us to read or see far away. [Ch 10: Light: Mirrors and Lenses]	1



Q.NO.	QUESTIONS	MARKS
13	Assertion (A): The Moon takes about 50 minutes longer than the previous day to appear in nearly the same position in the sky. Reason (R): While the Earth completes one rotation in 24 hours, the Moon moves ahead in its orbit around the Earth during this time. (A) A is true but R is false. (B) A is false but R is true. (C) Both A and R are true, and R is the correct explanation of A. (D) Both A and R are true, but R is NOT the correct explanation of A. Answer: Correct Option: (C) ► Why? C. Both statements are true. The source states that the Moon takes about 50 minutes longer to come back to nearly the same position in the sky. The reason correctly explains this because as Earth rotates once in 24 hours, the Moon moves ahead in its orbit, so Earth must rotate some more for the Moon to appear in the same spot. [Ch 11: Keeping Time with the Skies]	1
14	Why are spices with salt or sugar added to food items like pickles? (A) They act as preservatives and do not allow microorganisms to grow (B) They reduce the nutritional value of food (C) They improve the color of the food (D) They make the food easier to cook Answer: Correct Option: (A) ► Why? A. Spices with salt or sugar act as preservatives because high concentration of salt or sugar does not allow microorganisms to grow on the food. [Ch 2: The Invisible Living World: Beyond Our Naked Eye]	1
15	Assertion (A): Human beings are not crushed under the atmospheric pressure. Reason (R): The pressure inside our bodies is equal to the atmospheric pressure, which balances the pressure from outside. (A) Both A and R are true, but R is NOT the correct explanation of A. (B) Both A and R are true, and R is the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (B) ► Why? B. The assertion is true because the source states that we are not crushed under atmospheric pressure. The reason correctly explains this by stating that the pressure inside our bodies equals atmospheric pressure, balancing the external pressure. [Ch 6: Pressure, Winds, Storms, and Cyclones]	1



Q.NO.	QUESTIONS	MARKS
16	If Earth were much smaller than its actual size but had the same average density, what would happen to its atmosphere according to the chapter? (A) The atmosphere would have more oxygen for breathing (B) The gases in the atmosphere would escape into space due to weak gravity (C) The atmosphere would become thicker and denser (D) The atmosphere would remain exactly the same Answer: Correct Option: (B) ► Why? B. If Earth were much smaller with the same average density, its gravity would be too weak to hold on to the gases in the atmosphere, and they would escape into space. [Ch 13: Our Home: Earth, a Unique Life Sustaining Planet]	1
17	Assertion (A): Hydrogen and oxygen present in water cannot be separated by physical methods, but sodium chloride dissolved in water can be separated by the physical process of evaporation. Reason (R): The constituent elements of a compound cannot be separated by any physical method, but components of a mixture can be separated by physical processes. (A) A is false but R is true. (B) Both A and R are true, and R is the correct explanation of A. (C) A is true but R is false. (D) Both A and R are true, but R is NOT the correct explanation of A. Answer: Correct Option: (B) ► Why? B. Both statements are true according to the textbook. Water is a compound where hydrogen and oxygen are tightly attached and cannot be separated physically. Sodium chloride in water forms a mixture, and its components can be separated by evaporation. The reason correctly explains both parts of the assertion by stating the fundamental difference between compounds and mixtures. [Ch 8: Nature of Matter: Elements, Compounds, and Mixtures]	1
18	Why are there fewer dragonflies in ponds with fish compared to ponds without fish? (A) Fish eat dragonfly larvae (B) Dragonflies cannot fly near fish (C) Fish compete with dragonflies for food (D) Dragonflies avoid ponds with water Answer: Correct Option: (A) ► Why? A. Fish eat dragonfly larvae. Ponds with fish had fewer dragonflies because fish eat dragonfly larvae. [Ch 12: How Nature Works in Harmony]	1



Q.NO.	QUESTIONS	MARKS
19	In traditional Indian practice, variolation was used to protect against smallpox. What did this method involve? (A) Administering an inactivated bacterial toxin (B) Using material from a smallpox sore to scratch the skin (C) Injecting cowpox sap into a healthy person (D) Taking medicines called antibiotics Answer: Correct Option: (B) ► Why? B. Variolation involved using material from a smallpox sore to scratch the skin and create a mild infection to build immunity. People who performed this practice were known as teekedaars. [Ch 3: Health: The Ultimate Treasure]	1
20	Sample B (iron sulfide) is NOT attracted by a magnet because: (A) Iron sulfide contains no iron (B) Iron sulfide is too heavy to be attracted (C) Iron sulfide is a compound with completely different properties than its constituents (D) Iron sulfide is a liquid Answer: Correct Option: (C) ► Why? C. Iron sulfide is a compound formed by heating iron and sulfur. The new substance has completely different properties from its constituents, so it is not attracted by a magnet. [Ch 8: Nature of Matter: Elements, Compounds, and Mixtures]	1
Section B Attempt 5 out of 7.		
21	How does pressure difference affect air movement? Give examples of weather phenomena that result from different pressure differences. Answer: A small difference in pressure can result in a gentle breeze, while a stronger pressure difference can lead to strong winds and sometimes even cyclones. The concepts of force and pressure determine how air moves, and these pressure differences are connected to powerful weather events. ► Why? Air naturally flows from high-pressure regions toward low-pressure regions to equalize the atmospheric balance. A greater difference in pressure creates a stronger force, resulting in faster wind speeds that manifest as phenomena ranging from gentle breezes to severe cyclones. [Ch 1: Exploring the Investigative World of Science]	2
	OR State the location and function of the microorganism associated with pulse crops. Answer: The microorganism lives in the roots of pulse crops. It provides nutrients for the growth of these crops, which is crucial for their healthy development and successful yield.	



Q.NO.	QUESTIONS	MARKS
	► Why? Rhizobium bacteria inhabit root nodules to establish a symbiotic relationship with the plant. They convert atmospheric nitrogen into soluble compounds, enriching the soil and providing essential nutrients for the crop's development. [Ch 2: The Invisible Living World: Beyond Our Naked Eye]	
22	How is gravitational force different from magnetic force and electrostatic force in terms of being attractive or repulsive? Answer: Gravitational force is always an attractive force. In contrast, magnetic force and electrostatic force can either be attractive or repulsive depending on the nature of the interacting bodies or charges. ► Why? Mass exists in only one form, resulting in a force that is exclusively attractive, whereas electric charges and magnetic poles have opposite types that can repel each other. [Ch 5: Exploring Forces]	2
	OR State two reasons why nichrome wire is commonly used in electrical heating devices. Answer: Nichrome wire is commonly used in electrical heating devices because: (i) It is a good conductor of electricity, allowing current to flow through it. (ii) It generates more heat for a given current compared to other materials, making it efficient for heating applications like electric heaters and irons. ► Why? Nichrome possesses high resistivity, which ensures significant heat generation when current flows, and it has a high melting point that prevents the wire from melting under high temperatures. These properties allow it to efficiently convert electrical energy into heat while remaining physically stable. [Ch 4: Electricity: Magnetic and Heating Effects]	
23	Why is it advised to keep doors and windows open during storms with high-speed winds? Answer: When wind moves over the roofs and through the houses, the pressure difference between the inside of the houses and over the roofs is reduced to a large extent. This helps prevent the roofs from being blown off during high-speed winds, making it safer to keep doors and windows open. ► Why? High-speed winds blowing over a roof create a low-pressure zone, while the air inside the house remains at a higher pressure. Opening windows allows air to flow through, balancing these pressures and preventing the strong upward force that can blow the roof off. [Ch 6: Pressure, Winds, Storms, and Cyclones]	2
24	Fill in the blanks: (i) The solution used in a Voltaic cell is called ____. (ii) A current carrying coil behaves like a ____. Answer: (i) electrolyte (ii) magnet	2



Q.NO.	QUESTIONS	MARKS
	► Why? (i) The chapter explains that a Voltaic cell contains two metal plates and a liquid called an electrolyte, placed in a glass or plastic container. The electrolyte is usually a weak acid or salt solution. (ii) As stated in the chapter, a current carrying coil that behaves as a magnet is called an electromagnet. When electric current flows through a coil, it produces a magnetic field around it. [Ch 4: Electricity: Magnetic and Heating Effects]	
25	Why do antibiotics not work against viral infections? Answer: Antibiotics target parts of bacterial cells that are different from human cells, killing only bacteria. They do not work against viruses because viruses have different structures than bacteria. They are also ineffective against diseases caused by protozoa. ► Why? Antibiotics function by targeting specific cellular machinery found in bacteria, such as cell walls, which are absent in viruses. Since viruses are non-cellular and have a completely different structure, the drugs have no specific target to disrupt. [Ch 3: Health: The Ultimate Treasure]	2
Section C Attempt 5 out of 7.		
26	What is antibiotic resistance? Explain how it develops in bacteria and list three precautions to prevent its spread. Answer: Antibiotic resistance is a phenomenon where bacteria that were earlier killed by a given antibiotic survive and multiply despite treatment with that antibiotic. This makes common infections harder to treat and increases the risk of complications, prolonged illness, and even death. Resistance develops when a few microbes develop resistance to antibiotics. When antibiotics kill harmful bacteria, they also kill good bacteria protecting the body. The antibiotic-resistant bacteria then grow, take over, and can transfer resistance to other bacteria. To prevent spread: 1. Use antibiotics only when prescribed by a doctor 2. Take the correct dose for the right duration 3. Avoid unnecessary use, including not giving antibiotics to cattle unless required	3



Q.NO.	QUESTIONS	MARKS
	► Why? Resistance arises through natural selection, where bacteria with genetic mutations survive treatment and pass this trait to their offspring. Misuse of antibiotics accelerates this process by eliminating susceptible bacteria and allowing resistant strains to thrive. Adhering to precautions ensures these drugs remain effective by limiting opportunities for resistant populations to grow and spread. ► Step-by-step: 1. Step 1: Define antibiotic resistance - Bacteria survive and multiply despite treatment with an antibiotic that earlier killed them. 2. Step 2: Explain development - A few microbes develop resistance, antibiotics kill both harmful and good bacteria, and resistant bacteria grow and transfer resistance to others. 3. Step 3: State consequences - Makes infections harder to treat, increases risk of complications and death. 4. Step 4: List precautions - Use antibiotics only when prescribed, in correct dose, for right duration, and avoid unnecessary use including in animals. [Ch 3: Health: The Ultimate Treasure]	
	OR Explain the basic structure of a cell and describe the functions of the cell membrane and cytoplasm. Answer: Cells have three main parts: a thin outer lining called the cell membrane, a central region, and a small round structure called the nucleus. The cell membrane encloses the cytoplasm and nucleus, separates one cell from another, and is porous to allow entry of materials essential for life processes and exit of waste material. The cytoplasm fills the space between the cell membrane and nucleus. It contains other cell components and compounds such as carbohydrates, proteins, fats, and mineral salts. Most life processes take place within the cytoplasm. ► Why? Cells are organized into distinct regions where the membrane acts as a selective barrier controlling movement, and the cytoplasm provides the medium for organelles and metabolic reactions. This response correctly identifies the three main structural components and links them to their specific roles in maintaining life processes. ► Step-by-step: 1. Step 1: Recall that cells consist of three main parts: the cell membrane, cytoplasm, and nucleus. 2. Step 2: Explain the function of the cell membrane as a porous barrier that separates cells and controls the movement of materials. 3. Step 3: Describe the cytoplasm as the region containing essential compounds where most life processes occur. [Ch 2: The Invisible Living World: Beyond Our Naked Eye]	
27	The weight of an object on the Moon becomes one-sixth of its weight on the Earth. What causes this change? Does the mass of the object also become one-sixth of its mass on the Earth?	3



Q.NO.	QUESTIONS	MARKS
	Answer: The weight of an object on the Moon is one-sixth of its weight on Earth because the Moon's gravitational force is much weaker than Earth's gravitational force. Weight is the gravitational force with which a planet pulls an object, and different planets have different gravitational forces. However, the mass of the object does not change. Mass is the amount of matter in an object and remains the same everywhere. The mass stays at its original value, not one-sixth, because mass is independent of gravitational force. ► Why? The chapter provides a table showing that a 1 kg object weighs 10 N on Earth but only 1.6 N on the Moon. It explains that 'mass is the amount of matter in an object... Its value remains the same at every place. Weight, on the other hand, is the gravitational force... Since gravitational force can vary... weight can change, but mass does not.' ► Step-by-step: 1. The weight on the Moon is one-sixth of the weight on Earth due to the Moon's weaker gravitational force. 2. Weight is defined as the gravitational force with which a planet pulls an object. 3. Different planets have different gravitational forces, causing weight to vary. 4. Mass is the amount of matter in an object and remains constant everywhere. 5. Mass does not become one-sixth because it is independent of gravitational force. [Ch 5: Exploring Forces]	
	OR What are interparticle attractions? Explain how the strength of these attractions affects the arrangement and movement of particles in solids. Answer: Interparticle attractions are forces that hold the constituent particles of matter together. These forces are attractive in nature. The strength of these attractions depends on the nature of the substance and the interparticle distance. Even a slight increase in the distance decreases the interparticle forces drastically. In solids, the particles are tightly packed and the interparticle attractions are very strong. These strong forces hold the particles in fixed positions, preventing them from moving freely. The particles can only vibrate or oscillate about their positions but cannot move past each other. This arrangement gives solids their definite shape and volume.	



Q.NO.	QUESTIONS	MARKS
	► Why? Interparticle attractions function as the binding forces that hold particles together. In solids, these attractions are strong enough to lock particles into fixed positions, limiting their movement to slight vibrations rather than free motion. This strong binding force is the reason solids possess a rigid structure with a definite shape and volume. ► Step-by-step: 1. Step 1: Define interparticle attractions as attractive forces holding constituent particles together. 2. Step 2: Explain that strength depends on nature of substance and interparticle distance. 3. Step 3: Describe how in solids, particles are tightly packed with very strong attractions. 4. Step 4: Conclude that strong forces keep particles in fixed positions, allowing only vibration, giving solids definite shape and volume. [Ch 7: Particulate Nature of Matter]	
28	A boy lies horizontally on a loose sand bed. The same boy then stands vertically on the same sand bed. In which case will he sink more? Explain your answer with proper reasoning. Answer: The boy will sink more when he stands vertically on the sand bed. When the boy lies horizontally, his entire body is in contact with the sand, creating a large surface area. When he stands vertically, only his feet touch the sand, creating a much smaller contact area. Since the boy's weight remains the same in both cases, the pressure exerted on the sand is much greater when standing. This is because when the same force acts on a smaller area, the pressure increases. Greater pressure causes the boy to sink deeper into the sand when standing compared to when he lies horizontally. Pressure is inversely proportional to the area of contact when the applied force (the boy's weight) remains constant. Since standing reduces the contact area drastically compared to lying down, the pressure exerted on the sand increases significantly, causing him to sink deeper. ► Why? Pressure is inversely proportional to the area of contact when the applied force (the boy's weight) remains constant. Since standing reduces the contact area drastically compared to lying down, the pressure exerted on the sand increases significantly, causing him to sink deeper. ► Step-by-step: 1. Step 1: Compare the contact area in both cases - lying horizontally creates a larger contact area, while standing creates a smaller contact area. 2. Step 2: Recognize that the boy's weight (force) remains the same in both positions. 3. Step 3: Understand that when the same force acts on a smaller area, the pressure is greater. 4. Step 4: Conclude that greater pressure when standing causes more sinking in the sand. [Ch 6: Pressure, Winds, Storms, and Cyclones]	3
29	How do pressure differences lead to different weather conditions? Explain with examples.	3



Q.NO.	QUESTIONS	MARKS
	Answer: Pressure refers to how force is distributed over an object. The concepts of force and pressure determine how air moves. A small difference in pressure can result in a gentle breeze, while a stronger pressure difference can lead to strong winds, and sometimes even cyclones. These forces are connected to powerful weather events like storms and cyclones that affect our daily lives, agriculture, and even our safety. Thus, the magnitude of pressure difference directly determines the intensity of weather phenomena we experience. ► Why? Air naturally flows from high-pressure areas to low-pressure areas, creating wind as it attempts to balance the atmospheric conditions. The intensity of the resulting weather depends on the magnitude of this difference, as larger pressure gaps drive faster winds and more severe storms like cyclones. ► Step-by-step: 1. Step 1: Define pressure as how force is distributed over an object. 2. Step 2: Explain that small pressure differences result in gentle breeze. 3. Step 3: Explain that stronger pressure differences lead to strong winds and cyclones. [Ch 1: Exploring the Investigative World of Science]	
30	How does inserting an iron nail inside the paper cylinder affect the magnetic effect produced by a current-carrying coil, and what observation confirms this? Answer: When an iron nail is inserted inside the paper cylinder of a current-carrying coil, the magnetic effect increases significantly. This can be observed by placing magnetic compasses near the coil - the deflection in the compass needles becomes greater compared to when no nail was inserted. Additionally, when iron paper clips are placed near the two ends of the nail, the clips get attracted to the ends of the nail. The iron nail acts as a core that strengthens the magnetic field produced by the current-carrying coil. This arrangement demonstrates how an electromagnet works, where the magnetic effect can be controlled by switching the current on or off.	3



Q.NO.	QUESTIONS	MARKS
	► Why? Iron acts as a ferromagnetic core that concentrates the magnetic field lines, significantly increasing the magnetic strength compared to an air core. This stronger field exerts a greater force on nearby objects, resulting in the observed increase in compass deflection and the ability to attract items like paper clips. ► Step-by-step: 1. Step 1: The iron nail inserted in the cylinder acts as a core that enhances the magnetic field of the coil. 2. Step 2: The increased magnetic effect is observed by greater deflection in compass needles placed near the coil. 3. Step 3: Paper clips placed near the nail ends get attracted, confirming the stronger magnetic effect. [Ch 4: Electricity: Magnetic and Heating Effects]	

Section D
Attempt 4 out of 6.

31	What is pressure? Explain with a numerical example how pressure is calculated. How does the area over which a force is applied affect the pressure? Give two examples from daily life to support your answer. Also explain with an activity how liquids exert pressure. Answer: Pressure is defined as the force acting per unit area. It is calculated using the formula: $\text{Pressure} = \text{Force}/\text{Area}$. For example, if a force of 100 N is applied on a cardboard of area 2 m², then the pressure applied on the cardboard will be: $\text{Pressure} = 100 \text{ N}/2 \text{ m}^2 = 50 \text{ N/m}^2$. The area over which a force is applied has a significant effect on the resulting pressure. When the area is smaller, the resulting pressure is higher, making it easier to perform certain tasks. This is why it is easier to drive a nail using its pointed end rather than the head of the nail, because the pointed end has a smaller area, resulting in higher pressure. Similarly, it is easier to cut an apple with the sharp edge of a knife rather than the blunt edge, because the sharp edge has a smaller area, producing higher pressure. Liquids also exert pressure, which can be demonstrated through a simple activity. Take two transparent pipes of the same length (about 25 cm) but different diameters. Attach rubber balloons to one end of each pipe and clamp them on a stand. Fill both pipes with water up to the same level, about halfway. Observe that both balloons bulge to the same extent, even though the weight of water in the two pipes is different due to their different diameters. This shows that the pressure exerted by a liquid depends on the height of the water column, not on the weight of the water. Equal water column heights produce equal pressure, causing equal bulges in the balloons.	5
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Q.NO.	QUESTIONS	MARKS
	► Why? Pressure measures the concentration of force over a specific area, where reducing the contact area increases the pressure exerted for the same amount of force. This inverse relationship explains why sharp objects penetrate easily and how fluids exert force based on their weight and depth. ► Step-by-step: 1. Step 1: Define pressure as force per unit area and write the formula: $\text{Pressure} = \text{Force}/\text{Area}$. 2. Step 2: Provide a numerical example showing calculation: 100 N applied on 2 m² area gives pressure = 50 N/m². 3. Step 3: Explain the relationship between area and pressure - smaller area results in higher pressure. 4. Step 4: Give two daily life examples: driving a nail using pointed end and cutting an apple with sharp edge of knife. 5. Step 5: Describe the activity with two pipes and balloons to show that liquids exert pressure based on water column height, not weight of water. [Ch 6: Pressure, Winds, Storms, and Cyclones]	
	OR Explain the process of investigating a phenomenon like a scientist. Using the example of a puri puffing up in hot oil, describe the key steps involved and list the factors that can be changed or controlled during such an investigation. Answer: Investigation in science means more than just looking at something and asking simple questions. It involves asking focused questions, designing simple experiments to answer those questions, and using observations to improve understanding. Science begins with wonder, with simple "Why?" and "How?" questions about the world around us. The key steps in investigating like a scientist are: First, start with curiosity and careful observation. Ask "what happens if...?" questions. Then, find out two main things—what all can be changed or controlled when doing the experiment, and what all can be observed to see if these changes made any difference. Using the puri example: When a puri puffs up in hot oil, a scientist would observe carefully and ask why it puffs up like a balloon, and why one side is thinner than the other. To investigate this, we identify factors we can change or control: (1) the thickness of the rolled dough, (2) the size of the rolled dough, (3) different types of flour such as atta or maida, (4) the temperature of the hot oil, and (5) the way the rolled dough is dropped into the oil—vertically, at an angle, or slowly. These are things that we can control. The scientific principles that guide investigation are observing, measuring, and experimenting. We don't need a fancy laboratory—even a kitchen at home is a wonderful place to observe and ask questions. All we need is curiosity, careful observation, and asking "what happens if...?" Science is everywhere!	



Q.NO.	QUESTIONS	MARKS
	► Why? Scientific investigation requires moving from simple observation to a structured process of identifying variables that affect an outcome. The puri example illustrates this method by showing how one tests specific factors, such as oil temperature or dough thickness, to understand the underlying cause of the phenomenon. ► Step-by-step: 1. Step 1: Define what investigation in science means and how it goes beyond simple observation. 2. Step 2: Explain the key steps: starting with curiosity, asking focused questions, and identifying what can be changed versus what can be observed. 3. Step 3: Introduce the puri example and the kinds of questions a scientist would ask about it puffing up. 4. Step 4: List all the factors that can be changed or controlled: thickness, size, type of flour, temperature, and method of dropping. 5. Step 5: Conclude by mentioning the scientific principles (observing, measuring, experimenting) and that science can be done anywhere, even in the kitchen. [Ch 1: Exploring the Investigative World of Science]	
32	Forces can produce different effects on objects. What are the various effects that a force can have on the bodies on which it is applied? Explain each effect with a suitable example and classify the action as push or pull. Answer: A force is experienced as a push or pull in our daily lives, often without us even realising it. When a force is applied on a body, it can produce different effects depending on the situation. The various effects of force are: 1. Stopping or decreasing the speed of a moving object: When a force is applied, it can stop a moving object or reduce its speed. For example, when your friend holds your moving bicycle from behind to stop it, a pull is applied that stops or decreases the speed of the bicycle. 2. Changing the direction of a moving object: A force can change the direction in which an object is moving. For example, when a moving ball is hit with a bat, the push applied changes the direction of the moving ball. 3. Changing the shape of an object: When force is applied to certain objects, it can cause a change in their shape. For example, when an inflated balloon is pressed, the push causes a change in the shape of the balloon. Classification of forces as push or pull: • Actions like hitting a moving ball with a bat and pressing an inflated balloon involve a push. • Actions like holding a moving bicycle from behind to stop it involve a pull. It is important to analyse each situation to understand what kind of force (push or pull) is being applied and what effect it produces on the object.	5



Q.NO.	QUESTIONS	MARKS
	► Why? Force is defined as a push or pull that causes a change in an object's state of motion or its physical dimensions. This response correctly identifies the three primary effects: altering speed, changing direction, or deforming shape, which are the fundamental outcomes of applying force to any body. ► Step-by-step: 1. Step 1: Define force as a push or pull experienced in daily life. 2. Step 2: List the three main effects of force: stopping or changing speed, changing direction, and changing shape. 3. Step 3: Explain each effect with a specific example from everyday situations - bicycle stopping, ball direction change, and balloon shape change. 4. Step 4: Classify each example action as push or pull based on the nature of force applied. 5. Step 5: Conclude by emphasising the importance of analysing force situations to understand their effects. [Ch 5: Exploring Forces]	
	OR Describe an activity using a coil, an electric cell, and magnetic compasses to demonstrate how electric current affects magnetic materials. Explain the observations when the current is switched on and off, and how inserting an iron nail inside the coil changes the results. Answer: To demonstrate the effect of electric current on magnetic materials, take around 100 cm long flexible insulated wire, a piece of chart paper, an iron nail, an electric cell, two magnetic compasses, and few iron/steel paper clips. Roll the chart paper to make a cylinder of diameter roughly equal to the width of a pencil and secure it with adhesive tape. Tightly wind around 50 turns of the insulated wire on the cylinder to form a cylindrical coil and secure the wire with adhesive tape. Place the compasses near the two ends of the cylindrical coil. Connect the two ends of the coil with the terminals of the cell. Observe that the needles of the compasses show deflection. When the wire is disconnected from the cell, the needles return to their original positions, indicating the effect stops when current stops. Insert an iron nail in the paper cylinder and repeat the steps. There is a difference in the deflection of the compass needles compared to without the nail. When iron paper clips are placed near the two ends of the nail, the clips are attracted to the ends of the nail and cling to it. However, when the current is stopped, the clips no longer cling to it. Draw a diagram showing the cylindrical coil with 50 turns of insulated wire wound around the paper cylinder, the two ends of the coil connected to the electric cell, magnetic compasses placed near both ends of the coil showing deflection, and the iron nail inserted inside the cylinder with paper clips clinging to its ends. Label all parts clearly.	



Q.NO.	QUESTIONS	MARKS
	► Why? Electric current flowing through a coil generates a magnetic field, which deflects the compass needles due to the interaction of magnetic forces. When the current is switched off, the magnetic field collapses, causing the needles to realign with the Earth's magnetic field. Inserting an iron nail strengthens this magnetic effect because the ferromagnetic material concentrates the magnetic field lines within the coil. ► Step-by-step: 1. Step 1: List the materials required: 100 cm flexible insulated wire, chart paper, iron nail, electric cell, two magnetic compasses, and iron/steel paper clips 2. Step 2: Describe the construction of the coil by rolling chart paper into a pencil-width cylinder and winding 50 turns of insulated wire tightly around it 3. Step 3: Explain the observation of compass needle deflection when the coil is connected to the cell and how needles return to original position when disconnected 4. Step 4: Describe the effect of inserting an iron nail inside the coil on the deflection of compass needles and the attraction of paper clips to the nail ends 5. Step 5: Draw and label a diagram showing the coil with connecting wires, electric cell, compasses showing deflection, iron nail, and attracted paper clips [Ch 4: Electricity: Magnetic and Heating Effects]	
33	Explain what non-communicable diseases are with suitable examples. How do lifestyle-related diseases develop, and what steps can help prevent them? Also, describe the role of immunity in protecting the body from diseases. Answer: Non-communicable diseases are diseases that cannot spread from one person to another. These include deficiency diseases, which are caused by a lack of specific nutrients in the diet, and chronic diseases such as cancer, diabetes, and asthma. Chronic diseases may often persist for a long time, typically more than 3 months. Lifestyle-related diseases develop due to unhealthy habits and choices. For example, diabetes is becoming more prevalent in both adults and children. India now has one of the highest numbers of people with diabetes in the world. It develops due to a combination of factors: hormonal imbalances, unhealthy eating habits, lack of physical activity, being overweight or obese, and other reasons. The signs and symptoms of diabetes include frequent urination, excessive thirst, weight loss, tiredness, and slow healing. Prevention of lifestyle-related diseases involves making positive changes. For obesity, suggested lifestyle changes include eating a balanced diet and exercising regularly. The principle 'Prevention is better than cure' is important for protecting ourselves from both communicable and non-communicable diseases. Immunity is the natural ability of our body to fight diseases. Our body has a special system called the immune system that helps fight against diseases. Additionally, vaccines help prevent serious infections caused by viruses and bacteria. Vaccines protect us from diseases such as polio, measles, tetanus, and hepatitis. Simple steps like good sanitation, as seen in community campaigns, can also greatly reduce the spread of diseases and improve health.	5



Q.NO.	QUESTIONS	MARKS
	► Why? Non-communicable diseases are defined by their non-infectious nature, stemming from internal factors like genetics or deficiencies rather than external pathogens. A complete response links the development of lifestyle diseases to modifiable habits and explains how the immune system acts as the body's defense mechanism to maintain overall health. ► Step-by-step: 1. Step 1: Define non-communicable diseases and mention the two types - deficiency diseases and chronic diseases. 2. Step 2: Give examples of chronic diseases (cancer, diabetes, asthma) and explain that they persist for more than 3 months. 3. Step 3: Explain how lifestyle-related diseases like diabetes develop, listing the causes mentioned (hormonal imbalances, unhealthy eating, lack of physical activity, overweight/obesity). 4. Step 4: Describe the signs and symptoms of diabetes and mention lifestyle changes for prevention (balanced diet, regular exercise). 5. Step 5: Define immunity, describe the immune system, and explain how vaccines protect against diseases like polio, measles, tetanus, and hepatitis. [Ch 3: Health: The Ultimate Treasure]	
34	Describe the step-by-step procedure for preparing an onion peel slide for microscopic observation. Explain the purpose of each material used in the process and state what observations you would make when viewing the slide under a microscope. Answer: The preparation of an onion peel slide involves several systematic steps. First, take an onion bulb and wash it thoroughly with water. Cut the onion bulb vertically into pieces. Using forceps, pull out the thin, transparent layer from the inner surface of one piece. This layer is called the onion peel. Place the peel in a petri dish containing a few drops of safranin for 30 seconds. Safranin is a red-coloured stain that gives a pinkish colour to the cells, helping us see them clearly under the microscope. After staining, transfer the peel to another petri dish containing water using a thin brush to rinse the peel and remove extra stain. Carefully place the stained onion peel on a glass slide using a thin brush, ensuring it does not break or fold. Put a drop of glycerin over the peel. Glycerin prevents drying of the cells and improves clarity for better visualisation of cells. Slowly place a coverslip over the peel using a needle, ensuring no air bubbles get trapped. Use blotting paper to gently wipe off any extra glycerin around the edges. When viewed under a microscope or foldscope, you will observe nearly rectangular structures. These are the cells of the onion peel, which are closely arranged without any space between them. This activity demonstrates that plants are made up of cells.	5



Q.NO.	QUESTIONS	MARKS
	► Why? Staining the transparent peel creates necessary contrast to visualize cellular structures, while glycerin prevents the specimen from drying out during observation. Following these systematic steps ensures the delicate cells remain intact and distinct enough to identify components like the nucleus and cell wall. ► Step-by-step: 1. Step 1: Describe the initial preparation - washing the onion bulb, cutting it vertically, and removing the thin transparent layer (onion peel) using forceps. 2. Step 2: Explain the staining process - placing the peel in safranin (red-coloured stain) for 30 seconds to give cells a pinkish colour for clear visibility. 3. Step 3: Describe the rinsing step - transferring the peel to water using a thin brush to remove extra stain. 4. Step 4: Explain mounting the peel on glass slide with glycerin - glycerin prevents drying and improves clarity for better visualisation. 5. Step 5: Describe placing the coverslip carefully using a needle to avoid air bubbles and wiping extra glycerin with blotting paper. 6. Step 6: State the observations - nearly rectangular cells closely arranged without any space between them, showing plants are made up of cells. [Ch 2: The Invisible Living World: Beyond Our Naked Eye]	
Section E Attempt 3 out of 4.		
35	Case Study: Effects of Electric Current Ravi conducts an experiment for his science exhibition. He connects a nichrome wire to a cell and observes that the wire becomes warm after some time. He places a magnetic compass near the wire and notices the needle deflects from its original position. He then constructs an electromagnet by wrapping nichrome wire around an iron nail. He prepares three electromagnets with 25, 50, and 100 turns respectively to compare their strength. He also tests coils made of copper and aluminium wire to observe compass deflection. (i) Name the two effects of electric current demonstrated in Ravi's experiment. (ii) State one reason why nichrome is preferred over copper for making heating elements in electric devices. (iii) Predict how the lifting capacity of the electromagnet changes when the number of turns increases from 25 to 100. Explain your answer. (iv) Will the compass needle show deflection when placed near a coil made of aluminium wire carrying current? Justify your answer. Answer: (i) Heating effect and magnetic effect of electric current. (ii) Nichrome generates more heat for a given current (or Nichrome is cheaper than copper). (iii) The lifting capacity increases. The strength of the electromagnet increases with the number of turns of the coil. (iv) Yes. When current is passed through coils made of aluminium (and other metals like copper, iron, and nichrome), compass needles placed near the coils show deflection.	5



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	► Why? i) Heating effect and magnetic effect of electric current. ii) Nichrome generates more heat for a given current (or Nichrome is cheaper than copper). iii) The lifting capacity increases. The strength of the electromagnet increases with the number of turns of the coil. iv) Yes. When current is passed through coils made of aluminium (and other metals like copper, iron, and nichrome), compass needles placed near the coils show deflection. [Ch 4: Electricity: Magnetic and Heating Effects]	
	OR Case Study: Systematic Scientific Investigations Class 8 students are conducting systematic investigations to understand everyday phenomena. One group is studying how puris puff up by changing only the oil temperature while keeping dough thickness and dropping method identical. They record yes/no observations about puffing and measure the time in seconds it takes to puff. Another group examines how we use electric current in daily life—depending on its heating effect to keep warm and magnetic effect to run motors. A third group investigates health by exploring how tiny unseen microbes include both helpful ones that aid digestion and harmful ones that cause infections, and how medicines and vaccines help us stay healthy. (i) What specific numerical measurement does the first group record when studying how puris puff up? (ii) Why does the group use circles of dough of the same thickness and drop them in the same way when testing different oil temperatures? (iii) Identify the two effects of electric current mentioned in the investigation and state one practical use of each effect. (iv) Distinguish between helpful microbes and harmful microbes based on their effects on the human body. Answer: (i) The time in seconds it takes to puff up. (ii) To change only one thing at a time while keeping the other conditions same (for systematic investigation). (iii) Heating effect—used to keep us warm; Magnetic effect—helps motors run (and machines function). (iv) Helpful microbes aid digestion and produce medicines, while harmful microbes cause infections. ► Why? i) The time in seconds it takes to puff up. ii) To change only one thing at a time while keeping the other conditions same (for systematic investigation). iii) Heating effect—used to keep us warm; Magnetic effect—helps motors run (and machines function). iv) Helpful microbes aid digestion and produce medicines, while harmful microbes cause infections. [Ch 1: Exploring the Investigative World of Science]	



Q.NO.	QUESTIONS	MARKS
36	Case Study: Exploring the World of Microorganisms Riya, a Class 8 student, conducted experiments to study microorganisms. She collected pond water and observed it under a microscope that magnifies objects 100 to 400 times. She recorded seeing Amoeba with an irregular shape and Paramecium that moved using specialised structures. She also prepared a soil suspension from garden soil and observed tiny organisms in it. Later, she noticed a cotton-like growth on a tomato left outside for several days. When she asked why homemade pickles never showed such growth, her grandmother explained that adding salt and spices acts as preservatives. (i) Define microorganisms. (ii) Why is a microscope necessary to observe the organisms in pond water? (iii) Based on Riya's observations, identify the category of microorganisms for Amoeba and Paramecium. Write one characteristic of each that she recorded. (iv) Explain why pickles preserved with salt and spices do not get infected by microorganisms easily. Answer: (i) Microorganisms or microbes are tiny creatures that cannot be seen with the naked eye. (ii) A microscope is necessary because microorganisms cannot be seen with the unaided eye. The microscope magnifies them 100 to 400 times, making them visible. (iii) Both Amoeba and Paramecium belong to the category of Protozoa. Amoeba has an irregular shape. Paramecium moves with the help of specialised structures. (iv) High concentration of salt or sugar does not allow microorganisms to grow on them. Salt and spices act as preservatives that prevent microbial growth. ► Why? i) Microorganisms or microbes are tiny creatures that cannot be seen with the naked eye. ii) A microscope is necessary because microorganisms cannot be seen with the unaided eye. The microscope magnifies them 100 to 400 times, making them visible. iii) Both Amoeba and Paramecium belong to the category of Protozoa. Amoeba has an irregular shape. Paramecium moves with the help of specialised structures. iv) High concentration of salt or sugar does not allow microorganisms to grow on them. Salt and spices act as preservatives that prevent microbial growth. [Ch 2: The Invisible Living World: Beyond Our Naked Eye]	5



Q.NO.	QUESTIONS	MARKS
37	Case Study: Health and Disease Management During Health Week, students learned that health means complete physical, mental, and social well-being—not just the absence of disease. The doctor explained that infectious diseases are caused by pathogens like bacteria and viruses, while non-communicable diseases like diabetes result from lifestyle factors. She clarified that symptoms are what patients feel, whereas signs are visible or measurable. The program highlighted how vaccines train the immune system using dead or weakened germs, and emphasized that early diagnosis with continuous care helps control non-communicable diseases through lifestyle changes and medication. (i) What is the complete definition of health? (ii) Differentiate between 'symptoms' and 'signs' of a disease. (iii) Suggest two strategies for managing non-communicable diseases like diabetes to improve quality of life. (iv) Explain why vaccines are effective against infectious diseases but not against diabetes. Answer: (i) Complete physical, mental, and social well-being—not just the absence of disease. (ii) Symptoms are what we feel (like pain or tiredness), while signs are what can be seen or measured (like fever or rash). (iii) Managing symptoms through medication and improving quality of life through lifestyle changes and rehabilitation, ensuring continuous care to control disease progression. (iv) Vaccines train the immune system using dead, weakened, or harmless parts of a germ to prevent disease, but diabetes is a non-communicable disease caused by lifestyle and environmental factors, not by pathogens. ► Why? i) Complete physical, mental, and social well-being—not just the absence of disease. ii) Symptoms are what we feel (like pain or tiredness), while signs are what can be seen or measured (like fever or rash). iii) Managing symptoms through medication and improving quality of life through lifestyle changes and rehabilitation, ensuring continuous care to control disease progression. iv) Vaccines train the immune system using dead, weakened, or harmless parts of a germ to prevent disease, but diabetes is a non-communicable disease caused by lifestyle and environmental factors, not by pathogens. [Ch 3: Health: The Ultimate Treasure]	5

